INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

TEST STAND, AIRCRAFT GENERATOR UNITS

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID is for a constant speed drive (CSD) test stand for field testing multi-aircraft integrated-drive generators (IDG), CSD generators, and their associated generators.

2. SALIENT CHARACTERISTICS.

2.1 Equipment. A current and complete test stand shall be provided. The proposed test stand shall be of modular design and shall include all items and accessories required for testing all the identified IDGs and CSDs (see table II). When actual accuracies or specified ranges are not directly specified, the "best commercial practice" shall be proposed.

2.2 Function. Field-level test stand is used to test aircraft IDGs; the IDG consists of the CSD unit and its associated generator. The test stand shall accept and successfully operate all the IDG systems shown in table II and the applicable components of the adapter kits in table I. The test stand shall have its own oil cooler(s) and shall be capable of maintaining the CSD/IDG oil temperature required without the aid of the existing MC-2 or other auxiliary oil coolers. The auxiliary oil coolers in the MC-2 adapter kits shall not be used, nor are they required.

2.3 Commercial non-developmental item (NDI). The test stand shall be an already-developed, state-of-the-art, market-proven, commercial test stand with a proven reliability track record. Minor modifications are allowed to the existing test stand. Modifications, such as rearrangement or mounting of units on skid frame and changes to interface the unique military MC-2 adapters for IDG and CSD units, are considered acceptable NDI modifications.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: WR-AFLCMC/WN-SEV, 235 Byron St., Suite 19A, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4920

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TABLE I. Adapter kits.

Application	Model No.	Part No.	NSN
B-52, KC- 135B, EC/RC-135C	FSAK-3008	688307A	4920-01-147-0432
C-135	FSAK-3330	65997	4920-00-532-2713
C-135B	FSAK-3327	693581/A	4920-00-512-3432
F-4C	FSAK-3333	697466	4920-00-795-9698
F-4C	FSAK-3341	701484	4920-00-929-6805
F-15	FSAK-3351	712667	4920-00-171-6621
E-3A/RC-135M/W	FSAK-3359	715734A	4920-01-013-1998
F-16	FSAK-3360	718301	4920-01-024-5099
T-38/F-5	None	9520-0007	4920-01-486-1313

2.3.1 Major modules of assemblies. If the proposed test stand is assembled using commercial NDI major modules or assemblies, which have not been previously merged and sold commercially as a test stand, the major modules or assembly that is proposed to be used shall have a proven reliability track record of reliable and satisfactory service in similar test stands. Offeror shall provide evidence that these are proven and reliable, commercially-available assemblies.

2.3.2 Ambient-temperature ranges.

Test Stand during Operation:	0 ºF	to	125 ºF
Load Bank Operation (outdoor environment):	-25 ºF	to	130 ºF
Storage, All Units:	-65 ºF	to	160 ºF

2.4 Description. The proposed test stand shall be a commercial NDI test stand. It shall be compactly packaged, arranged, and suitable for testing all the identified CSDs and IDGs. The test stand shall not require any modification of the “real property” (test-cell floor space area, walls, ceiling, or utility/power support systems) used with the current MC-2 test stands.

2.4.1 Digital-entry devices. All inputs and control settings shall use digital keypads or digital-keyboard entry devices. The entry keypad(s) and/or keyboard(s) shall be rugged, environmentally-sealed devices for use in wet, dirty, or extreme environments. Toggle or selector switches are allowed for selecting the desired load on the load bank; however, the total selected load(s) must be indicated on electronic digital display(s).

2.4.2 Electronic digital displays. Except for protective alarms, all indicators shall be electronic digital displays. Electronic digital displays shall be light-emitting diode (LED), liquid-crystal display (LCD), vacuum-fluorescent, or electro-fluorescent type devices. Mechanical counters, odometers, or turn-counter devices are not acceptable electronic digital displays.

2.4.3 Data input and output consistent formats. Display formats should be consistent within a system. When appropriate for users, the same format should be used for input and output. Data entry formats should match the source document formats.

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TABLE II: Constant speed drive systems.

	CSD System										GEN
Aircraft	Model No	P/N	Tech. Manual 9H6-Series	Oil-in Testing Temp. (°F)	ΔT Max (°F)	Flow Rate in GPM	Input Speed (Rated) (RPM)	Load Testing Input Speed (RPM) (±50 - ±100)	Accel/ Decel rpm/sec (±50)	Charge Pressure (PSI)	Test Load 3 ϕ kW
B-52H	120AGD04	708754A	5-5-2	150±10 125±10	50	12	4475 - 7980 ccw	2500 - 7900 **	800 max	240 - 500	54
KC-135B	"	"	"	"	"	"	"	"	"	"	"
EC/RC-135C	"	"	"	"	"	"	"	"	"	"	"
KC-135	40CSD30	691680A	3-15-2	200±10	40		3600 - 7000 ccw	3650 - 7000	800 max	280 - 380	36
C-135B	40AD18	694256B	3-23-2	200±10	35	5.7	4475 - 8814 ccw	4500 - 8814	800 max	300 - 360	39
F-4C	30ADG05	695145G	3-25-2	185±20 200±10		6.5	4750 - 7685 ccw	4900 - 7685		275 - 500	30
E-3A		714672M	*	200±10	40	6.75	4565 - 8765 ccw	4800 - 8600		220 - 260	43
RC-135M/W		727410A	*	200±10	40	6.75	4593 - 7987 ccw	4800 - 8600		220 - 260	43
F-16A/B		734556	3-44-3	250±10 200±10 150±10	45	16.5	5000 - 9000 ccw	5000 - 9000	500 and 950	240 - 300	60
F-16C/D		734556A	3-44-3	250±10 200±10 150±10	45	16.5	5000 - 9000 ccw	5000 - 9000	500 and 950	240 - 300	60
F-15		711595L 711595J	3-36-2	200±10	30	16.5	4500 - 9000 ccw	4600 - 8000	800 max	240 - 300	30
F-15E		739827A 739827B	5-14-2	200±10	40		5000 - 8750 ccw	4375 - 7000	100 and 500	240 - 320	30

* T.O. 9H23-5-2

** 3 kW load required for underspeed testing.

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2.4.4 Labeling application. Labeling shall be of a consistent type and consistently located within the entire system. Labels, legends, placards, signs, markings, or a combination of these shall be provided whenever personnel must identify items (except when it is obvious to the observer what an item is).

2.4.5 Major modules of assemblies. While the final configuration is dependent on the commercial NDI test stand proposed, some of the major modules or assemblies should consist of:

Control Console Drive
Drive Control Panel
Gearbox
Hydraulic Supply
Oil Cooler
Interface Panel
Load Bank
Electrical Hoist
Mounting Pad for CSD Adapters
Work Bench
Instrumentation as Required
Unit/Prime Mover
Drive Power Supply
Gearbox Output pad
Hydraulic Instrumentation Panel
Oil Heater
Load Bank Control Panel
Air-cooling Blower
Skid (Mounting) Frame
Blast-air Blower
Built-in Digital Multi-meter

2.4.6 Mounting pad for CSD adapters. A single mounting pad shall be provided for permitting compatibility with the mechanical adapters currently used with the current MC-2 test stands.

2.4.7 Interface panel. An interface panel shall be provided as the electrical connection points for the generator under the test and CSD adapter, and associated generator control units/voltage regulators. The connecting points, connectors, test leads, and indicators in the interface panel shall be compatible with all the existing MC-2 adapters.

2.4.8 Drive unit. The drive unit shall consist of a 125 to 150 horsepower (hp), direct current (DC) drive motor. The controller shall have a state-of-the-art, microprocessor-based, DC power supply with electronic digital inputs and outputs. The power supply shall contain built-in self-test capabilities and shall have protection against drive unit overspeed for safe operation of the drive unit and the unit under test (UUT).

2.4.9 Load bank. The load bank shall be a 75 to 90 kilowatt (kW), 400 hertz (Hz) (PF 1.0) resistive load bank. To allow for outdoor heat dissipation, the load bank shall be in a rain-tight and splash-proof enclosure. The incremental loading steps shall be selected to allow operator 1 kW resolution loading for the entire range of the load bank. Short-time (1 to 5 minutes) maximum loading shall permit loading up to 150% of the continuous rating of the load bank. Electronic digital display(s) shall indicate the total selected load. The load bank shall have built-in, protection-safe operation of the load bank. At least 50 feet of cabling shall be provided for the remote outdoor location of the load bank. The controls and metering for the remotely located display shall be in the proximity of the main control panel. Load bank controls shall include selection of the load steps, master load switch, and blower-air fail indicator.

2.4.10 Blast-air blower. A built-in, cooling air blower with all necessary adapters and flexible hoses shall be provided for generators that require blast-air cooling. Air pressure (0-16 inches of water) and volume shall be adjusted as required.

2.4.11 Hydraulic oil cooler. The hydraulic oil cooler shall be provided with 10-micron nominal supply and return filters with pre-heater and heat exchanger, quick-disconnect couplings, and

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automatic temperature and pressure controls. UUT protective alarms to shut down the drive system for flow-loss and over-temperature shall be provided. The oil cooler shall have means for providing external pressure for priming or purging the UUT (drive units) prior to running tests. A pressure regulator with gauges shall be provided. The hydraulic oil cooler pressure required is 0 to 15 pounds per square inch (psi) maximum.

2.4.12 Work bench. A work bench shall be provided for test equipment, adapters, voltage regulators, etc. The work bench shall be rugged and capable of safely holding at least 150 pounds of equipment.

2.4.13 Built-in digital multi-meter. A multi-meter shall be provided with appropriate probes and test leads for testing and troubleshooting.

2.5 Specifications. All speeds and power requirements shall be met using all the voltage and frequencies listed in table III.

TABLE III. Requirements.

Hydraulic system operating oil	MIL-PRF-7808 or MIL-PRF-23699
Hydraulic oil flow rate	20 gpm maximum capability
Oil reservoir capacity	12 gallons (approximate)
Hoist capacity	0 – 500 lb
Output power: continuous operation	125 to 150 hp (in speed range: 4200 to 11,000 rpm)
Output power: intermittent 1-minute operation	187 hp (in speed range: 4200 to 11,000 rpm)
Speed drive	Electronically-controlled variable 0 – 11,000 rpm
Acceleration/deceleration rate	Adjustable 50 to 1200 rpm/sec
Output shaft rotation	Clockwise and counter-clockwise
Instrumentation	
RPM indicator	0 – 15,000 rpm
Kilowatt meter range	0 – 125 kW, PF 1.0
AC voltage range	0 – 300 VAC
DC voltage range	0 – 30 VDC
AC ammeter range	0 – 250 amps (as required)
Frequency counter range	0 – 999.9 Hz
Test timer	0 – 99 min 59 sec
Lapse timer	9999.9 hr
Charge oil-pressure gage	0 – 600 psi
Oil-temperature controller	50 – 400 °F
Oil-temperature gage	0 – 500 °F
Input power	220/380/440 VAC, 3-phase, 50/60 Hz
Length	110 in (approximate)
Width	80 in (approximate)
Height	80 in (approximate)
Weight	7,000 lbs (approximate)

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2.6 Disturbance to other elements. The equipment shall be constructed to permit the replacement and adjustment of components and accessories with minimum disturbance to and without removal of other elements for the units.

2.7 Electrical connectors and quick-disconnect fastenings. The test stand shall include the maximum use of electrical connectors and quick-disconnect fastenings to allow for rapid removal and replacement of components parts. Covers or access plates that must be removed for component adjustments or for component or parts removal shall be equipped with durable quick-disconnect fastenings or combinations of quick-disconnect fastenings and hinges.

2.8 Special test equipment. In the event that any special test equipment is required for the operation or maintenance of this test stand, the special test equipment shall be identified and it shall be included with the offer. Contractor failure to identify and include required special test equipment within the offer is considered grounds for contract default.

2.9 First article.

(1) First article inspection and limited tests are required at contractor site (see 4.3).

(2) First article inspection and complete performance and compatibility tests are required at a government site (see 4.4).

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or biobased materials. The contractor is encouraged to use recovered materials or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403.

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale commercially. The Government reserves the right to require proof of such conformance.

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to request that the contractor provide proof of such conformance prior to first delivery, and thereafter, as may be appropriate.

4.3 First article inspection and test at contractor site. The contractor shall perform the Phase 1, First Article Inspection and Tests, prior to delivery. Approved Phase 1 Test Procedure delivered per contract data requirements list (CDRL) A002, DI-NDTI-80603A (Test Procedure) shall be used. A Phase 1 Test Report per CDRL A003, DI-NDTI-80809B (Test/Inspection Report) is required. The government shall witness all inspections and testing.

4.4 First article testing at government site. The contractor shall perform the Phase II, first article testing at a government site to be identified after contract award. Approved Phase II Test Procedure delivered per CDRL A002, DI-NDTI-80603A (Test Procedure) shall be used. A Phase II Test Report per A003, DI-NDTI-80809B (Test/Inspection Report) is required. The government shall witness all inspections and testing. The testing shall demonstrate that the test stand can successfully run and test

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all the CSD systems listed in this document and table II. First article testing at a government site does not relieve the contractor from ensuring that the test meets all performance requirements. All defects of failures will be corrected and satisfactorily demonstrated by the contractor prior to final acceptance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order. In the absence of specific instructions in the contract or order, the manufacturer's standard commercial packaging and marking practices will be used.

6. NOTES.

6.1 Intended use. These test stands will be used for field testing multi-aircraft IDGs, CSDs, and their associated generators.

6.2 Associated data item descriptions.

6.2.1 First article Test Procedures; DI-NDTI-80603A, CDRL A002, data item 01.

6.2.2 First article Test/Inspection Report; DI-NDTI-80809B, CDRL A003, data item 02.

6.3 Required modification. The technical proposal should include drawings, sketches, and other technical data to describe the modifications required for this application.

6.4 Market proven/acceptance. Offeror should be providing proof of market acceptance and testing of the offered unit by providing test data, performance history, warranty repair data, and/or reliability test data. Also, submit commercial literature, performance specifications, specification sheets, receipts or invoices, and lists of customers with addresses and telephone numbers to show the offered test stand has a proven, reliable history (track record). Evidence that the offered test stand has been sold as a complete test stand in significant quantities for some time is highly desirable.

6.5 Evaluation data. Proposals will be evaluated for compliance with section 3 of this CID. Bidders should submit the following data with their offer for evaluation.

6.5.1 Part number and manufacturer of offered item(s).

6.5.2 Part number or model number and photographs of the test stand that will be modified.

6.5.3 Provide a side-by-side comparison of how the proposed units meet or will meet all the minimum essential criteria in sections 2 thru 2.9 of this CID.

6.5.4 Provide contractor certifications of meeting all the published and proposed technical specifications and requirements.

6.5.5 Provide evidence that the proposed unit is reliable and commercially "market proven" per 2.3 and 2.3.1.

6.5.5.1 Any available data or evidence providing reliability.

6.5.5.2 Descriptive literature for commercial test stand (commercial catalog cut-sheets, specification sheets, etc.).

6.5.5.3 Operation & Maintenance manuals for the commercial test stand that will be modified.

6.5.5.4 Location of a commercially-similar test stand and point of contact for the government to inspect if deemed necessary by the evaluation team.

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6.6 Source of documents.

6.6.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks listed below form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

DI-NDTI-80603	-	Test Procedure
DI-NDTI-80809	-	Test/Inspection Report
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167

(Copies of these documents are available at https://assist.dla.mil.)

DEPARTMENT OF DEFENSE DOCUMENTS

DOD 5010.12-M	-	PROCEDURES FOR THE ACQUISITION AND MANAGEMENT OF TECHNICAL DATA

(Copies of these documents are available at http://www.dtic.mil/whs/directives/index.html.)

6.6.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however supersedes applicable laws and regulations.

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MILITARY INTEREST

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Preparing Activity:
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